

Markovova lokálnosť

Uprav na začiatku →

0	1	2	3
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1. Model snímača relatívnej polohy σ_k

min. 3 rovnice

$$P(\Delta p = \sigma_k - 2) = 0,05$$

$$P(\Delta p = \sigma_k - 1) = 0,2$$

$$P(\Delta p = \sigma_k) = 0,5$$

$$P(\Delta p = \sigma_k + 1) = 0,2$$

$$P(\Delta p = \sigma_k + 2) = 0,05$$

2. Model snímača absolútnej polohy (presnejší)

$$\begin{matrix} = 0 \\ \uparrow \end{matrix}$$

$$P(p = a_k - 2) = 0$$

$$P(p = a_k - 1) = 0,1$$

$$P(p = a_k) = 0,8$$

$$P(p = a_k + 1) = 0,1$$

$$P(p = a_k + 2) = 0$$

3. Overdepodobnosť polohy robota v predchádzajúcom kroku p_{k-1} a info zo snímačov / príkaz

$$P(p_{k-1} = 0) = 0$$

$$P(p_{k-1} = 1) = 0,15$$

$$P(p_{k-1} = 2) = 0,7$$

$$P(p_{k-1} = 3) = 0,15$$

Príkaz $p_k = 3$ potom ak $p_{k-1} = 2$ teda $\Delta p = +1$

Absolútny snímač potvrdil $p_k = 3 \rightarrow \underline{a_k = 3}$

Relatívny snímač potvrdil $p_k = 2 \rightarrow$ potom $\underline{\Delta p = 2 - 2 = 0}$

$$\underline{\sigma_k = \Delta p = 0}$$

0	1	2	3
		• →	
		σ_k	a_k

4. Počítame P pre každú polohu p_k v $L = \{0, 1, 2, 3\}$ ak

$$\begin{aligned} P(p_k) &= P(\text{absolútneho modelu}) \cdot P(\text{relatívneho modelu}) \\ &= P(a_k | p_k) \cdot \sum [P(\sigma_k, p_{k-1} | p_k) \cdot P(p_{k-1})] \end{aligned}$$

$$\underline{P(n_k=0)} = \underbrace{P(a_k=3 | n_k=0)}_{\text{čokolicek}} = 0$$

$$P(n_k = a_k - 3) = 0!$$

$$\underline{P(n_k=1)} = \underbrace{P(a_k=3 | n_k=1)}_{\text{čokolicek}} = 0$$

$$P(n_k = a_k - 2) = 0!$$

$$\underline{P(n_k=2)} = \underbrace{P(a_k=3 | n_k=2)}_{P(n_k=a_k-1)=0,1} \cdot [P(n_k=2, n_{k-1}=0 | \sigma_k=0) \cdot P(n_{k-1}=0) +$$

$$+ P(n_k=2, n_{k-1}=1 | \sigma_k=0) \cdot P(n_{k-1}=1) +$$

$$+ P(n_k=2, n_{k-1}=2 | \sigma_k=0) \cdot P(n_{k-1}=2) +$$

$$+ P(n_k=2, n_{k-1}=3 | \sigma_k=0) \cdot P(n_{k-1}=3)]$$

$$= 0,1 \cdot [\overset{2-0=2}{\underset{\uparrow}{P(\Delta n = \sigma_k + 2)}} \cdot 0 + \overset{2-1=1}{\underset{\uparrow}{P(\Delta n = \sigma_k + 1)}} \cdot 0,15 + \overset{2-2=0}{\underset{\uparrow}{P(\Delta n = \sigma_k)}} \cdot 0,7 +$$

$$+ \underset{\downarrow}{P(\Delta n = \sigma_k - 1)} \cdot 0,15]$$

$$= 0,1 \cdot (\overset{2-3=-1}{\underset{\downarrow}{0,2}} \cdot 0,15 + 0,5 \cdot 0,7 + 0,2 \cdot 0,15) = 0,1 \cdot (0,03 + 0,35 + 0,03) = 0,1 \cdot 0,45 =$$

$$= \underline{0,045}$$

$$\underline{P(n_k=3)} = \underbrace{P(a_k=3 | n_k=3)}_{P(n_k=a_k)=0,8} \cdot [P(n_k=3, n_{k-1}=0 | \sigma_k=0) \cdot P(n_{k-1}=0) +$$

$$+ P(n_k=3, n_{k-1}=1 | \sigma_k=0) \cdot P(n_{k-1}=1) +$$

$$+ P(n_k=3, n_{k-1}=2 | \sigma_k=0) \cdot P(n_{k-1}=2) +$$

$$+ P(n_k=3, n_{k-1}=3 | \sigma_k=0) \cdot P(n_{k-1}=3)]$$

With respect to electric energy

VONSCH
power electronics

$$\begin{aligned}
 &= 0,8 \cdot \left[\overset{3-0=3}{P(\Delta \uparrow = \sigma_k + 3)} \cdot \overset{=0}{0} + \overset{3-1=2}{P(\Delta \uparrow = \sigma_k + 2)} \cdot 0,15 + \overset{3-2=1}{P(\Delta \uparrow = \sigma_k + 1)} \cdot 0,7 + \right. \\
 &\quad \left. + \underset{\substack{\downarrow \\ 3-3=0}}{P(\Delta \uparrow = \sigma_k)} \cdot 0,15 \right] \\
 &= 0,8 \cdot [0,05 \cdot 0,15 + 0,2 \cdot 0,7 + 0,5 \cdot 0,15] = 0,8 \cdot (0,0075 + 0,14 + 0,075) = \\
 &= 0,8(0,2225) = \underline{\underline{0,178}}
 \end{aligned}$$

5. Normalizácia

$$\Sigma = 0 + 0 + 0,041 + 0,178 = 0,219$$

$$\underline{\underline{P(\uparrow_k = 2)}} = \frac{0,041}{\Sigma} = \underline{\underline{0,187}}$$

$$\underline{\underline{P(\uparrow_k = 3)}} = \frac{0,178}{\Sigma} = \underline{\underline{0,812}}$$